

QUESTION 2.

Evaluate the following indefinite integrals.

(a) $\int \frac{\sec x}{1 + \sin x} dx$. (Hint: You may consider the substitution $u = \sin x$.)

(b) $\int \frac{\sqrt{x^2 + 4x - 5}}{x + 2} dx$.

AY2223 Q2

a) Let $u = \sin x \Rightarrow du = \cos x dx$

$$\int \frac{\sec x}{1 + \sin x} dx = \int \frac{\sec x}{1 + u} \cdot \frac{1}{\cos x} du$$

$$= \int \frac{1}{1+u} \cdot \frac{1}{\cos^2 x} du$$

$$= \int \frac{1}{1+u} \cdot \frac{1}{1-u^2} du$$

$$= \int \frac{1}{(1-u)(1+u)^2} du$$

$$\frac{1}{(1-u)(1+u)^2} = \frac{A}{1-u} + \frac{B}{1+u} + \frac{C}{(1+u)^2}$$

$$= \frac{1}{4(1-u)} + \frac{1}{4(1+u)} + \frac{1}{2(1+u)^2}$$

$$\therefore = \frac{1}{4} \int \frac{1}{1-u} du + \frac{1}{4} \int \frac{1}{1+u} du + \frac{1}{2} \int \frac{1}{(1+u)^2} du$$

$$= \frac{1}{4} \frac{\ln|1-u|}{-1} + \frac{1}{4} \ln|1+u| + \frac{1}{2} \frac{(1+u)^{-2+1}}{-2+1} + C$$

$$= \frac{1}{4} \ln \left| \frac{1-u}{1+u} \right| - \frac{1}{2(1+u)} + C$$

$$= \frac{1}{4} \ln \left| \frac{1 + \sin x}{1 - \sin x} \right| - \frac{1}{2(1 + \sin x)} + C$$

b) $\int \frac{\sqrt{x^2 + 4x - 5}}{x+2} dx = \int \frac{\sqrt{(x+2)^2 - 9}}{x+2} dx$ $\cos \theta = \frac{3}{x+2}$

Let $x+2 = 3 \sec \theta \Rightarrow dx = 3 \sec \theta \tan \theta d\theta$

$$\int \frac{\sqrt{(x+2)^2 - 9}}{x+2} dx = \int \frac{\sqrt{9 \sec^2 \theta - 9}}{3 \sec \theta} \cdot 3 \sec \theta \tan \theta d\theta$$

$$= \int 3 \sqrt{\sec^2 \theta - 1} \cdot \tan \theta d\theta$$

$$= \int 3 \tan \theta \tan \theta d\theta$$

$$= 3 \int \tan^2 \theta d\theta$$

$$= 3 \int (\sec^2 \theta - 1) d\theta$$

$$= 3 [\tan \theta - \theta] + C$$

$$= 3 \left[\frac{\sqrt{x^2 + 4x - 5}}{3} - \cos^{-1} \left(\frac{3}{x+2} \right) \right] + C$$



Question 3. Determine whether each of the following series converges or diverges. Justify your answer. (20 marks)

$$(a) \sum_{n=1}^{\infty} \frac{2n^2 + 3n}{\sqrt{n^7 + 1}}$$

$$(b) \sum_{n=1}^{\infty} \tan\left(\frac{n\pi}{4n + \sqrt{n}}\right)$$

$$(c) \sum_{n=1}^{\infty} \frac{1}{4^n - 3^n}$$

$$\sum a_n \quad \text{(Limit)} \quad \text{(20 marks)}$$

$$\textcircled{1} a_n = \frac{\text{polynomial}}{\text{poly}} = \frac{\text{poly}}{\text{poly}} \rightarrow \text{Comparison test using p-series}$$

$$\textcircled{2} a_n = \frac{c^n \pm d^n}{d^n \pm e^n} \rightarrow \text{(Limit)} \quad \text{Comparison Test using Geometric Series}$$

$$\textcircled{3} a_n \text{ combinations of poly, } c^n, n! \rightarrow \text{Ratio Test}$$

$$\textcircled{4} \text{ Divergence Test}$$

$$\boxed{a_n \not\rightarrow 0} \Rightarrow \sum a_n \text{ diverges}$$

$$\Rightarrow a_n \text{ contains } (-1)^n \Rightarrow \text{Try ratio, root, AST}$$

$$\textcircled{5} a_n \text{ contains trig functions.} \Rightarrow \text{try comparison.}$$

Look at their uncalculus series first.

$$\textcircled{6} \text{ If can integrate } a_n \text{ (e.g. } a_n \text{ contains } \ln n) \Rightarrow \text{try integral test}$$

AY1718 Q3

a) $0 < a_n = \frac{2n^2 + 3n}{\sqrt{n^3 + 1}} \leq \frac{2n^2 + 3n}{\sqrt{n^3}} \leq \frac{2n^2 + 3n^2}{\sqrt{n^3}} = \frac{5n^2}{\sqrt{n^3}} = \frac{5}{n^{1/3}} = b_n$

Since $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{5}{n^{1/3}}$ converges (p-series $p = 1.5 > 1$), by comparison Test, $\sum_{n=1}^{\infty} a_n$ converges.

b) Let $a_n = \tan\left(\frac{n\pi}{4n+3}\right)$.

$$\begin{aligned} \lim_{n \rightarrow \infty} a_n &= \tan\left(\lim_{n \rightarrow \infty} \frac{n\pi}{4n+3}\right) \\ &= \tan\left(\lim_{n \rightarrow \infty} \frac{\pi}{4+\frac{3}{n}}\right) \\ &= \tan\left(\frac{\pi}{4}\right) = 1 \neq 0 \end{aligned}$$

By Divergence Test, $\sum a_n$ diverges

c) $a_n = \frac{1}{4^n - 3^n}$ $b_n = \frac{1}{4^n}$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{a_n}{b_n} &= \lim_{n \rightarrow \infty} \frac{1}{4^n - 3^n} \bigg/ \frac{1}{4^n} \\ &= \lim_{n \rightarrow \infty} \frac{1}{1 - \left(\frac{3}{4}\right)^n} \\ &= 1 \end{aligned}$$

Since $\sum b_n = \sum \frac{1}{4^n}$ Converges, by Limit Comparison Test,

$\sum a_n$ converges.

$$\sum c_n (x-a)^n$$

(20 marks)

*a = 5 centre***Question 5.**

- (a) Find the interval of convergence of the following power series, and identify the values of x for which the series converges absolutely or conditionally. Justify your answer.

$$\sum_{n=0}^{\infty} \frac{n^6 (x-5)^n}{n^8 + 1}.$$

- (b) If the series $\sum_{n=1}^{\infty} a_n$ converges absolutely, does the series $\sum_{n=1}^{\infty} a_n^2$ converge? Justify your answer.

AY1718 Q5

a) Idea: apply Ratio Test.

• deal with end points case by case

$$\text{let } a_n = \frac{n^6(x-5)^n}{n^3+1}$$

$$\begin{aligned} \rho &= \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)^6(x-5)^{n+1}}{(n+1)^3+1} \cdot \frac{n^3+1}{n^6(x-5)^n} \right| \\ &= |x-5| \lim_{n \rightarrow \infty} \frac{(n+1)^6}{n^6} \cdot \frac{n^3+1}{(n+1)^3+1} \\ &= |x-5| \lim_{n \rightarrow \infty} \frac{(1+\frac{1}{n})^6}{1} \cdot \frac{1+\frac{1}{n^3}}{(1+\frac{1}{n})^3+1} \\ &= |x-5| \frac{(1+0)^6}{1} \cdot \frac{1+0}{(1+0)^3+0} \\ &= |x-5| \end{aligned}$$

By Ratio Test,

• the series converges (absolutely) if

$$\begin{aligned} \rho &< 1 \text{ i.e. } |x-5| < 1 \\ -1 &< x-5 < 1 \\ 4 &< x < 6 \end{aligned}$$

• the series diverges if

$$x > 6, x < 4$$

Consider end points

$x=4$ The series becomes

$$\sum_{n=0}^{\infty} \frac{n^6(-1)^n}{n^3+1}$$

$$\text{let } a_n = \frac{n^6(-1)^n}{n^3+1}$$

$$0 < |a_n| = \frac{n^6}{n^3+1} < \frac{n^6}{n^3} = \frac{1}{n^3}$$

By comparison Test, since $\sum \frac{1}{n^3}$ converges, $\sum (a_n)$ converges.
 $\sum a_n$ converges absolutely.

$x=6$ The series becomes

$$\sum_{n=1}^{\infty} \frac{n^6}{n^3+1}$$

This converges absolutely by the same argument above.

b) We know $\sum a_n$ converges absolutely

i.e. $\sum |a_n|$ converges

$$\Rightarrow |a_n| \rightarrow 0$$

$|a_n| \rightarrow 0$ means $\exists N$ s.t.

$$|a_n| < 1 \text{ for } n > N$$

$$0 \leq |a_n| < 1 \text{ for } n > N$$

Multiply by $|a_n|$:

$$0 \leq |a_n|^2 < |a_n| \text{ for } n > N$$

By comparison Test, $\sum_{n=N}^{\infty} a_n^2$ converges

So $\sum_{n=1}^{\infty} a_n^2$ converges

Question 6.

(20 marks)

- (a) Find a power series with center $c = 2$ that represents the function $f(x) = \frac{1}{4 + 3x}$.
- (b) Find the first five nonzero terms of the Maclaurin series for $f(x) = \frac{1}{1 + \sin x}$.
- (c) Using power series or otherwise, find the sum $\sum_{n=0}^{\infty} n e^{-n}$.

$$\begin{aligned} \text{a) } \frac{1}{4+3x} &= \frac{1}{4+3(x-2+2)} \\ &= \frac{1}{10+3(x-2)} \\ &= \frac{1}{10} \cdot \frac{1}{1+\frac{3}{10}(x-2)} \\ &= \frac{1}{10} \cdot \frac{1}{1-(-\frac{3}{10}(x-2))} \\ &= \frac{1}{10} \sum_{n=0}^{\infty} \left[\left(-\frac{3}{10}\right)(x-2) \right]^n \end{aligned}$$

$$\text{I.O.C: } \left| -\frac{3}{10}(x-2) \right| < 1$$

$$|x-2| < \frac{10}{3} \rightarrow \mathbb{R}$$

$$\boxed{2 - \frac{10}{3} < x < 2 + \frac{10}{3}} \rightarrow \text{I.O.C}$$

$$\begin{aligned} \text{b) } \frac{1}{1+\sin x} &= \frac{1}{1-(-\sin x)} \\ &= \sum_{n=0}^{\infty} (-\sin x)^n \\ &= 1 - \sin x + (\sin x)^2 - (\sin x)^3 + (\sin x)^4 - \dots \\ &= 1 - \left(x - \frac{x^3}{3!} + \dots\right) + \left(x - \frac{x^3}{3!} + \dots\right)^2 - \left(x - \frac{x^3}{3!} + \dots\right)^3 + \left(x - \frac{x^3}{3!} + \dots\right)^4 - \dots \end{aligned}$$

constant: 1

coef. of x : -1

coef. of x^2 : 1

coef. of x^3 : $\frac{1}{3!} - 1 = -\frac{5}{6}$

coef. of x^4 : $-\frac{1}{4!} + 2 + 1 = \frac{2}{3}$

$1 - x + x^2 - \frac{5}{6}x^3 + \frac{2}{3}x^4 + \dots$

These are the first 5 non-zero terms required

$$\text{c) } \sum_{n=0}^{\infty} n e^{-n}$$

Idea: identify suitable expression for x

identify suitable power series to manipulate

$$\sum_{n=0}^{\infty} n e^{-n} = \sum_{n=0}^{\infty} n \left(\frac{1}{e}\right)^n$$

maybe should think $\sum_{n=0}^{\infty} n x^n$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

$$\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} n x^{n-1}$$

$$\frac{x}{(1-x)^2} = \sum_{n=1}^{\infty} n x^n$$

$$\sum_{n=1}^{\infty} n \left(\frac{1}{e}\right)^n = \frac{\frac{1}{e}}{(1-\frac{1}{e})^2}$$

Question 3. Determine whether each of the following series converges or diverges. Justify your answer. (15 marks)

(a) $\sum_{n=1}^{\infty} \ln \left(\frac{n^4 + 2n^3}{2n^4 + n^3} \right).$

(b) $\sum_{n=1}^{\infty} \frac{7^n + n^2}{8^n - n}.$

(c) $\sum_{n=1}^{\infty} n!e^{-n^2}.$

AY1819 Q3

$$\begin{aligned} \text{a) } \lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} \ln \left(\frac{n^4 + 2n^3}{2n^4 + n^3} \right) \\ &= \ln \left(\lim_{n \rightarrow \infty} \frac{n^4 + 2n^3}{2n^4 + n^3} \right) \\ &= \ln \left(\lim_{n \rightarrow \infty} \frac{1 + \frac{2}{n}}{2 + \frac{1}{n}} \right) \\ &= \ln \left(\frac{1}{2} \right) \neq 0 \end{aligned}$$

By Divergence Test, $\sum a_n$ diverges.

$$\begin{aligned} \text{b) } \text{Let } a_n &= \frac{7^n + n^2}{8^n - n} \quad b_n = \frac{7^n}{8^n} \\ \lim_{n \rightarrow \infty} \frac{a_n}{b_n} &= \lim_{n \rightarrow \infty} \frac{7^n + n^2}{8^n - n} \bigg/ \frac{7^n}{8^n} \\ &= \lim_{n \rightarrow \infty} \frac{1 + \frac{n^2}{7^n}}{1 - \frac{n}{8^n}} \\ &= 1 > 0 \end{aligned}$$

Since $\sum b_n$ converges, by Limit Comparison Test, $\sum a_n$ converges.

$$\text{c) } \text{Let } a_n = n! e^{-n^2}$$

$$\begin{aligned} \rho &= \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \\ &= \lim_{n \rightarrow \infty} \frac{(n+1)! e^{-(n+1)^2}}{n! e^{-n^2}} \\ &= \lim_{n \rightarrow \infty} (n+1) e^{-n^2 - 2n - 1 + n^2} \\ &= \lim_{n \rightarrow \infty} (n+1) e^{-2n-1} \\ &= \lim_{x \rightarrow \infty} \frac{x+1}{e^{2x+1}} \\ &= 0 < 1 \end{aligned}$$

By Ratio Test, $\sum a_n$ converges absolutely.

Question 5.

(20 marks)

- (a) Find the interval of convergence of the following power series, and for every $x \in \mathbb{R}$, determine whether the series converges absolutely, converges conditionally or diverges at x . Justify your answer.

$$\sum_{n=2}^{\infty} \frac{(3x-7)^n}{n \ln n}$$

- (b) Let $\{a_n\}_{n=1}^{\infty}$ be a sequence such that $\lim_{n \rightarrow \infty} n^2 a_n = M$ for some real number $M \in \mathbb{R}$. Does the series $\sum_{n=1}^{\infty} a_n$ converge absolutely? Justify your answer.

$$n^2 a_n \rightarrow M \quad n^2 a_n \sim M$$

$$a_n \sim \frac{M}{n^2} \sim \frac{1}{n^2}$$

AY1819 Q5

b) Know that $n^2 a_n \rightarrow M$

So $\exists N$ s.t

$$|n^2 a_n - M| < 1 \text{ for } n > N$$

$$-1 < n^2 a_n - M < 1$$

$$-|M| - 1 < M - 1 < n^2 a_n < M + 1 < |M| + 1$$

$$|n^2 a_n| < |n| + 1$$

$$0 \leq |a_n| < \frac{|M| + 1}{n^2} \leq \frac{C}{n^2}$$

By Comparison Test, since $\sum \frac{C}{n^2}$ converges.

$\sum |a_n|$ converges

i.e. $\sum a_n$ converges absolutely.

Alternatively:

Choose $b_n = \frac{1}{n^2}$

In view of Limit Comparison Test:

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \frac{|a_n|}{b_n} = \lim_{n \rightarrow \infty} \frac{|a_n|}{1/n^2} \\ &= \lim_{n \rightarrow \infty} n^2 |a_n| \\ &= \left| \lim_{n \rightarrow \infty} n^2 a_n \right| \\ &= |M| \end{aligned}$$

Question 6.

(20 marks)

- (a) Find a power series with center $c = 1$ that represents the function $f(x) = \frac{1}{3-x}$.
- (b) Using the Maclaurin series, find the fourth derivative $f^{(4)}(0)$ at the point 0 of the function $f(x) = \sin(2x)\sqrt{1+x}$.
- (c) Using power series or otherwise, find the sum $\sum_{n=1}^{\infty} \frac{n}{(n+2)!}$.

AY1819 Q6

$$c) \sum_{n=1}^{\infty} \frac{n}{(n+2)!}$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\frac{e^x}{x^2} = \sum_{n=0}^{\infty} \frac{x^{n-2}}{n!}$$

$$= x^{-2} + x^{-1} + \frac{1}{2!} + \frac{x}{3!} + \frac{x^2}{4!} + \dots$$

$$\frac{d}{dx} \left(\frac{e^x}{x^2} \right) = -2x^{-3} - x^{-2} + \frac{1}{3!} + \frac{2x}{4!} + \dots$$

Multiply by x :

$$x \frac{d}{dx} \left(\frac{e^x}{x^2} \right) = -2x^{-2} - x^{-1} + \frac{x}{3!} + \frac{2x^2}{4!} + \frac{3x^3}{5!} + \dots$$

$$\underbrace{\sum_{n=1}^{\infty} \frac{n}{(n+2)!} x^n}$$

$$x \left(\frac{x^4 e^x - e^x (2x)}{x^4} \right) = -2x^{-2} - x^{-1} + \sum_{n=1}^{\infty} \frac{n x^n}{(n+2)!}$$

$$\sin x = 1$$

$$1 \left(\frac{e - 2e}{1} \right) = -2 - 1 + \sum_{n=1}^{\infty} \frac{n}{(n+2)!}$$

$$\sum_{n=1}^{\infty} \frac{n}{(n+2)!} = 3 - e$$

QUESTION 1.

(20 Marks)

- (a) Use the definition of limit to prove that

$$\lim_{n \rightarrow \infty} \frac{2n^2 - 3n}{3n^2 + 1} = \frac{2}{3}.$$

- (b) Suppose the sequence $\{a_n\}$ is convergent, where $a_n > 0$ for all n and $\lim_{n \rightarrow \infty} a_n = 4$.
Prove, by definition of limit, that

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{a_n}} = \frac{1}{2}.$$

- (c) Let $\{b_n\}$ and $\{c_n\}$ be two sequences where $\{b_n\}$ is divergent and $|c_n| \geq |b_n|$ for all n .
Is it true that $\{c_n\}$ is divergent? Justify your answer. Prove it if it is true, and give a counter-example if it is false.

AY2021 Q1

a) Show $\lim_{n \rightarrow \infty} \frac{2n^2 - 3n}{3n^2 + 1} = \frac{2}{3}$

Let $\varepsilon > 0$. W.T.F N.s.t

$$\left| \frac{2n^2 - 3n}{3n^2 + 1} - \frac{2}{3} \right| < \varepsilon$$

Working backwards:

$$\left| \frac{3(2n^2 - 3n) - 2(3n^2 + 1)}{3(3n^2 + 1)} \right| < \varepsilon$$

$$\Leftrightarrow \left| \frac{-9n - 2}{3(3n^2 + 1)} \right| < \varepsilon$$

$$\Leftrightarrow \frac{9n + 2}{3(3n^2 + 1)} < \varepsilon$$

$$\Leftrightarrow \frac{9n + 2}{3(3n^2)} < \varepsilon$$

$$\Leftrightarrow \frac{9n + 2n}{3(3n^2)} < \varepsilon \quad (n \geq 1)$$

$$\frac{11n}{9n^2} < \varepsilon$$

$$n > \frac{11}{9\varepsilon}$$

$$\text{Choose } N = \left\lceil \frac{11}{9\varepsilon} \right\rceil$$

b) Let $\varepsilon > 0$. W.T.F N.s.t

$$\left| \frac{1}{\sqrt{a_n}} - \frac{1}{2} \right| < \varepsilon$$

$$\Leftrightarrow \left| \frac{2 - \sqrt{a_n}}{2\sqrt{a_n}} \right| < \varepsilon$$

$$\Leftrightarrow \left| \frac{4 - a_n}{2\sqrt{a_n}(2 + \sqrt{a_n})} \right| < \varepsilon$$

$$\Leftrightarrow \left| \frac{4 - a_n}{2\sqrt{a_n}(2)} \right| < \varepsilon$$

consider the fact $a_n \rightarrow 4$

$$\exists K \text{ s.t. } |a_n - 4| < 3$$

$$-3 < a_n - 4 < 3$$

$$\underbrace{1 < a_n < 7}$$

$$\Rightarrow 1 < \sqrt{a_n}$$

$$\Rightarrow \frac{1}{\sqrt{a_n}} < 1 \text{ for } n > K$$

$$\Leftrightarrow \left| \frac{4 - a_n}{4} \right| < \varepsilon \quad \text{provided } (n > K)$$

$$\exists M \text{ s.t. } |a_n - 4| < 4\varepsilon \quad n > M$$

$$\Leftrightarrow \left| \frac{4 - a_n}{4} \right| = \frac{|4 - a_n|}{4} < \frac{4\varepsilon}{4} = \varepsilon$$

provided
 $n > M$

$$\text{Choose } N = \max \{K, M\}$$

QUESTION 4.

(16 Marks)

Determine whether each of the following series converges or diverges. Justify your answers.

(a) $\sum_{n=1}^{\infty} \frac{5^n}{6^n - n}$

(b) $\sum_{n=1}^{\infty} \frac{1}{2^{\ln n}}$

(c) $\sum_{n=1}^{\infty} \left(1 - \cos\left(\frac{1}{n}\right)\right)$ (Recall that $\sin x \leq x$ for $x \geq 0$.)

QUESTION 6.

(18 Marks)

- (a) Let $f(x) = (\cos x) \ln \sqrt{4 - x^2}$. Using Maclaurin series or otherwise, find the value of $f^{(4)}(0)$.
- (b) Determine the interval of convergence of the power series $\sum_{n=1}^{\infty} \frac{(-1)^n n}{5^n} x^{2n}$, and find the function it represents on this interval.